EE-170L Computer Programming Lab

Laboratory No 02

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i. Basic Syntax and Input and Output in C/C++
ii. Data Types, Indentations, Comments, and Arithmetic Operators in C++

Lab Objectives:

The goal of this lab is to familiarize students with the foundational elements of C++. By the end of this session, you will be able to set up a basic program structure, manage user input and output, apply correct data types, follow indentation standards, and perform calculations using arithmetic operators.

Basic Syntax, Input, and Output

Every C++ program follows a specific boilerplate structure. We use the iostream library to facilitate communication between the program and the user.

- **cout (Character Output):** Used to print data to the screen.
- **cin (Character Input):** Used to read data from the keyboard.
- **<< and >>:** These are stream insertion and extraction operators, respectively.

The Hello World & User Greeting

```
#include <iostream>
using namespace std;

int main() {
    // Basic Output
    cout << "Welcome to the C++ Lab!" << endl;

    string name;
    cout << "Enter your name: ";

    // Basic Input
    cin >> name;

    cout << "Hello, " << name << "! Let's start coding." << endl;

    return 0;
}
```

Data Types, Comments, and Operators

C++ is a statically-typed language, meaning you must declare the type of a variable before using it.

Common Data Types

Data Type	Description	Example
int	Integers (whole numbers)	int age = 20;
double	Floating-point numbers (decimals)	double price = 19.99;
char	Single characters	char grade = 'A';
bool	Boolean values	bool isPassed = true;
string	Sequence of characters	string text = "C++";

Arithmetic Operators

C++ uses standard symbols for math: + (Addition), - (Subtraction), * (Multiplication), / (Division), and % (Modulus/Remainder).

Best Practices: Indentation and Comments

- **Indentation:** Always indent code inside curly braces {} (usually 4 spaces or 1 tab) to make the logic readable.
- **Comments:** Use // for single-line notes and /* ... */ for multi-line explanations.

Simple Calculator

```
#include <iostream>
using namespace std;

int main() {
    // Variable Declaration
    int num1, num2;

    cout << "Enter two integers: ";
    cin >> num1 >> num2;

    // Performing Arithmetic Operations
    int sum = num1 + num2;
    int diff = num1 - num2;
    int product = num1 * num2;

    /* Using double for division to ensure
       decimal precision if needed */
    double quotient = (double)num1 / num2;
    int remainder = num1 % num2;

    // Displaying Results
    cout << "Sum: " << sum << endl;
    cout << "Difference: " << diff << endl;
    cout << "Product: " << product << endl;
    cout << "Quotient: " << quotient << endl;
    cout << "Remainder: " << remainder << endl;

    return 0;
}
```

Lab -02 Tasks:

1. **Declare a different type of variables and find its memory size, location and values.**

2. **Write a C++ program that declare and display a floating-point variable called ”pi” and assign it the value 3.14.**
3. **Write a C++ program that converts temperature from Celsius to Fahrenheit. Hint: $F = (C \times 9/5) + 32$.**
4. **Separate Task sheet will be provided to you for more tasks**